

Assessment: Java GUI – Creating a Simple Window (JFrame)

Instructions: Choose the letter of the correct answer.

1. What does GUI stand for?

- A. General User Instruction
- B. Graphical User Interface
- C. Global Unified Interface
- D. Graphic Utility Index

2. Which import statement is required to use the JFrame class, as shown in the video?

- A. `import java.util.JFrame;`
- B. `import javax.swing.JFrame;`
- C. `import java.awt.JFrame;`
- D. `import javax.swing.JFrame;`

3. In the video, what must the class do in order to gain JFrame's window-creating abilities?

- A. implement JFrame
- B. extends JFrame
- C. import JFrame as a variable
- D. call JFrame.create()

4. In the phrase 'public class Tutorial extends JFrame', what is JFrame considered in relation to Tutorial?

- A. JFrame is the subclass, Tutorial is the superclass
- B. JFrame is the superclass (parent class), Tutorial is the child class
- C. They are unrelated classes
- D. JFrame is an interface implemented by Tutorial

5. What is the purpose of the constructor in the Tutorial class in this example?

- A. To destroy the JFrame window
- B. To hold and run the window setup code (title, size, visibility, close operation) when an object is created
- C. To import the Swing library
- D. To declare the main method

6. What does the setTitle() method do?

- A. Sets the size of the window
- B. Sets the text that appears in the window's title bar
- C. Closes the program
- D. Changes the window's background color

7. Is setTitle() required for the JFrame window to function?

- A. Yes, without it the program won't compile
- B. No, it's optional, but it makes the window look nicer by showing a title in the bar
- C. Yes, it is required to set the window size
- D. No, it is only used for closing the window

8. Which method sets the window's dimensions, and what default size was used in the video?

- A. `setDimension(400, 400)`
- B. `setSize(400, 400)`
- C. `setBounds(400)`
- D. `setWindow(400x400)`

9. According to the video, can the user resize the window after it opens, even though a default size was set?

- A. No, the size is permanently fixed

- B. Yes, the user can still drag the corners to resize it
- C. Only the developer can resize it through code
- D. The window automatically resizes itself randomly

10. What does setVisible(true) do?

- A. Makes the window appear/show on the screen
- B. Deletes the window
- C. Changes the window's title
- D. Minimizes the window automatically

11. What would happen if setVisible(false) was used instead of setVisible(true)?

- A. The window would appear twice
- B. The window would not appear at all
- C. The window would appear but without a title
- D. The program would crash immediately

12. Why does the video say a boolean value is needed for setVisible()?

- A. Because Java needs to know whether the value is true or false to decide if the window should show
- B. Because booleans make the code run faster
- C. Because JFrame requires exactly two parameters
- D. Because visibility is measured in numbers, not true/false

13. What does setDefaultCloseOperation(EXIT_ON_CLOSE) do?

- A. It prevents the user from closing the window
- B. It makes sure the program fully terminates when the user clicks the X button
- C. It changes the size of the window when closed
- D. It hides the close button

14. What problem occurs if setDefaultCloseOperation(EXIT_ON_CLOSE) is NOT included in the program?

- A. The program will not run at all
- B. Clicking the X button will close the window visually, but the program will keep running in the background
- C. The title bar will disappear
- D. The window will not be resizable

15. In the main method, which line correctly creates an object of the Tutorial class to run the program, based on the video?

- A. Tutorial t = new Tutorial();
- B. new Tutorial.run();
- C. Tutorial.main(t);
- D. JFrame t = Tutorial();

16. According to the video, what is the advantage of using a constructor and creating an object, instead of writing all the setup code directly inside main()?

- A. It is required by Java syntax and has no other benefit
- B. It's faster and more organized, since creating one object automatically runs all the setup code in the constructor instead of repeatedly typing t.setTitle(), t.setSize(), etc.
- C. It prevents the window from closing
- D. It removes the need for the JFrame import

17. What is the correct method signature for the main method shown in the video?

- A. public void main(String args)
- B. public static void main(String[] args)
- C. static main(String args[])
- D. public main(void args)

18. What happens in Java's execution order according to the video: which runs first, the main method or the constructor?

- A. The constructor always runs first, then main
- B. Java reads the main method first; when the object is created there, it then runs the constructor
- C. They run at exactly the same time
- D. Java randomly decides which runs first

19. Which of these best lists, in order, the setup steps performed inside the constructor in the video?

- A. setVisible → setSize → setTitle → setDefaultCloseOperation
- B. setTitle → setSize → setVisible → setDefaultCloseOperation
- C. setDefaultCloseOperation → setTitle → setSize → setVisible
- D. setSize → setDefaultCloseOperation → setTitle → setVisible

20. What is the overall purpose of this tutorial, as summarized at the end of the video?

- A. To teach how to connect a Java program to a database
- B. To teach how to create a simple GUI window using JFrame, covering title, size, visibility, and close behavior
- C. To teach how to create console-based input programs only
- D. To teach advanced game development using Java

Answer Key

Item No.	Answer
1	B
2	B
3	B
4	B
5	B
6	B
7	B
8	B
9	B
10	A
11	B
12	A
13	B
14	B
15	A
16	B
17	B
18	B
19	B
20	B